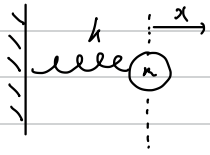




1. Simple Harmonic Motion.



Equation of motion:

$$ma = m\ddot{x} = -kx$$

$$\ddot{x} = -\frac{k}{m}x$$

$$\ddot{x} = -\omega^2 x, \quad \omega^2 = \frac{k}{m}$$

General solution: $x(t) = A \cos(\omega t + \phi_0)$

$A \rightarrow$ Amplitude

$\phi_0 \rightarrow$ Phase shift

$\omega \rightarrow$ Angular Frequency

$T \rightarrow$ Period

$\nu \rightarrow$ frequency

Relationship b/w T, ν, ω :

$$\omega = 2\pi/T = 2\pi\nu$$

$$\nu = 1/T = \omega/2\pi$$

$$T = \frac{2\pi}{\omega} = \frac{1}{\nu}$$

$$\text{OR: } x(t) = X_0 + A_1 \cos(\omega t) + A_2 \sin(\omega t)$$

$X_0 \rightarrow$ Offset, typically 0

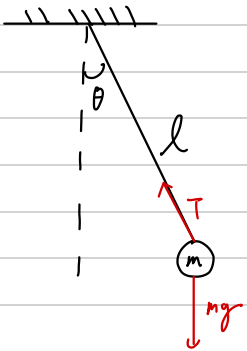
Energy of an oscillator:

$$\text{Total energy: } K + U = \frac{1}{2} k A^2$$

$$\begin{aligned} \text{Kinetic energy: } K &= \frac{1}{2} m v^2 = \frac{1}{2} m (-A\omega \sin(\omega t + \phi_0))^2 \\ &= \frac{1}{2} (m A^2 \omega^2 \sin^2(\omega t + \phi_0)) \end{aligned}$$

$$\text{Potential energy: } U = \frac{1}{2} k x^2 = \frac{1}{2} k A^2 \cos^2(\omega t + \phi_0)$$

Simple pendulum:



$$m l \frac{d^2 \theta}{dt^2} = -mg \sin \theta$$

angular acceleration

$\sin \theta \sim \theta$ for small θ

$$\frac{d^2 \theta}{dt^2} = -\frac{g}{l} \theta$$

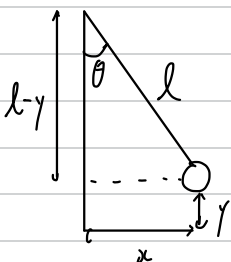
$$\text{or } x = A \sin\left(\sqrt{\frac{g}{l}} t + \phi\right)$$

where $x = l\theta$

$$\omega = \sqrt{\frac{g}{l}}$$

$$T = 2\pi \sqrt{\frac{l}{g}}$$

Energy of a simple pendulum:



$$1. \quad l^2 = (l-y)^2 + x^2$$

$$2ly = y^2 + x^2$$

$$\begin{aligned} 2. \quad &\text{Apply small angle approx.} \rightarrow y = \frac{x^2}{2l} \\ &x \ll l \Rightarrow y \ll x \\ &\therefore y^2 \sim 0 \end{aligned}$$

$$\text{Total energy} = mg \left(\frac{A^2}{2l}\right)$$

$$KE = \frac{1}{2} m v^2$$

$$= \frac{1}{2} m g \frac{A^2}{l} (A^2 - x^2)$$

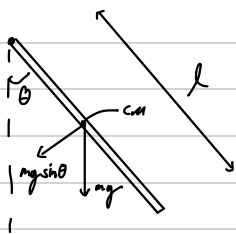
$$PE = \frac{1}{2} m g y$$

$$= \frac{1}{2} m g \frac{x^2}{2l}$$

$$= \frac{1}{4} \frac{m g x^2}{l}$$

$$\rightarrow x = A \sin\left(\sqrt{\frac{g}{l}} t + \phi\right)$$

Physical Pendulum:



$$I \frac{d^2\theta}{dt^2} = \tau$$

$$I \frac{d^2\theta}{dt^2} = -\frac{1}{2} m g \sin\theta$$

$$\rightarrow \frac{m l^2}{3} \frac{d^2\theta}{dt^2} = -\frac{1}{2} m g \sin\theta$$

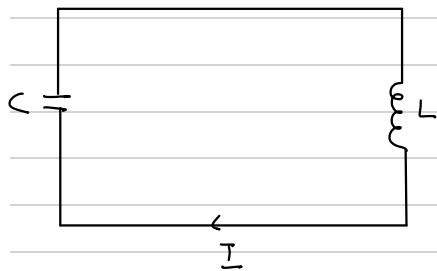
$$\frac{d^2\theta}{dt^2} = -\frac{3g}{2l} \theta, \omega = \sqrt{\frac{3g}{2l}}, T = 2\pi \sqrt{\frac{2l}{3g}}$$

Parallel axis theorem:

$$I_{\text{end}} = m d^2 + \int_{-\frac{l}{2}}^{\frac{l}{2}} \rho x^2 dx, \rho = \frac{m}{l}, \text{ uniform density}$$

$$\begin{aligned} \frac{l}{2} & \rightarrow \frac{m l^2}{4} + \frac{m l^2}{12} \\ & = \frac{m l^2}{3} \end{aligned}$$

LC Circuit:



Kirchoff's Voltage Law

$$V_C + V_L = 0$$

$$\frac{q}{C} + L \frac{dI}{dt} = 0 \quad \text{also} \quad I = \frac{dq}{dt}$$

$$L \frac{d^2q}{dt^2} = -\frac{q}{C}$$

$$\frac{d^2q}{dt^2} = -\frac{1}{LC} q, \omega = \sqrt{\frac{1}{LC}}, T = 2\pi \sqrt{LC}$$

For any SHM, $E = \frac{1}{2} a v^2 + \frac{1}{2} \beta x^2$

2. Damped Harmonic Oscillator

Damping force: $F_d = -bv$

Now, $mx'' = -kx - bv$

Equation of motion:

$$x'' + \frac{b}{m}x' + \frac{k}{m}x = 0 \quad , \quad x'' + \gamma x' + \omega_0^2 x = 0$$

$$\omega_0 = \sqrt{\frac{k}{m}}, \text{ natural frequency}$$

$$\gamma = \frac{b}{m}$$

Sol'n 1, Light damping:

$$x = A_0 \exp(-\gamma t/2) \cos \omega t, \text{ for } \omega_0^2 > \gamma^2/4, \quad \omega^2 = \omega_0^2 - \gamma^2/4$$

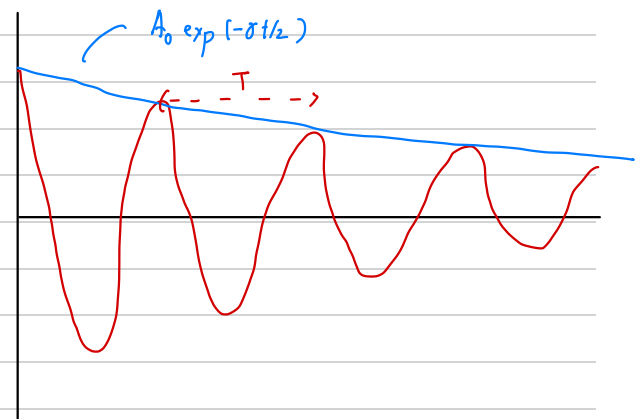
$$v = x' = -\gamma/2 A_0 \exp(-\gamma t/2) \cos \omega t - A_0 \omega \exp(-\gamma t/2) \sin \omega t$$

$$= -A_0 \exp(-\gamma t/2) \left(\omega \sin \omega t + \frac{\gamma}{2} \cos \omega t \right)$$

$$a = x'' = \frac{\gamma}{2} A_0 \exp(-\gamma t/2) (\omega \sin \omega t + \frac{\gamma}{2} \cos \omega t) - A_0 \exp(-\gamma t/2) (\omega^2 \cos \omega t - \omega \gamma \sin \omega t)$$

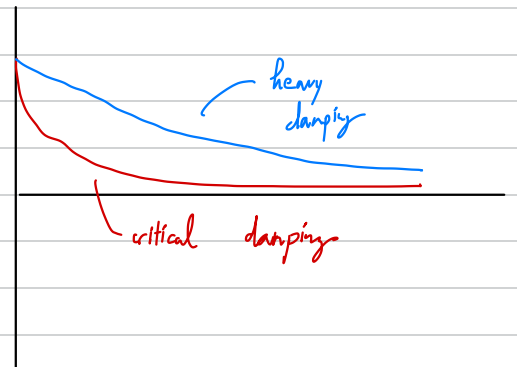
$$= A_0 \exp(-\gamma t/2) \left(\frac{\gamma}{2} \omega \sin \omega t + \frac{\gamma^2}{4} \cos \omega t - \omega^2 \cos \omega t + \frac{\gamma}{2} \omega \sin \omega t \right)$$

$$= A_0 \exp(-\gamma t/2) \left(\gamma \omega \sin \omega t + \left(\frac{\gamma^2}{4} - \omega^2 \right) \cos \omega t \right)$$



Sol'n 2, Heavy Damping

$$x = A \exp\left(-\frac{\gamma}{2} + \left(\frac{\gamma^2}{4} - \omega_0^2\right)^{1/2} t\right) + B \exp\left(-\frac{\gamma}{2} - \left(\frac{\gamma^2}{4} - \omega_0^2\right)^{1/2} t\right), \text{ for } \frac{\gamma^2}{4} > \omega_0^2$$



Sol'n 3, critical damping.

$$x = A \exp\left(-\frac{\gamma}{2} t\right) + B t \exp\left(-\frac{\gamma}{2} t\right)$$

Rate of energy loss for lightly damped oscillators
 $\hookrightarrow$ Assume $\omega_0^2 \gg \gamma^2/4$, $\omega \sim \omega_0$

$$E = K + U = \frac{1}{2} m v^2 + \frac{1}{2} k x^2$$

$$= \frac{1}{2} \left(m \left[A_0 \exp(-\gamma t/2) \left(\omega \sin \omega t + \frac{\gamma}{2} \cos \omega t \right) \right]^2 + k \left[A_0 \exp(-\gamma t/2) \cos \omega t \right]^2 \right)$$

$$= \frac{1}{2} \left(m A_0^2 \exp(-\gamma t) \left(\omega^2 \sin^2 \omega t + \gamma \omega \sin \omega t \cos \omega t + \frac{\gamma^2}{4} \cos^2 \omega t \right) + k A_0^2 \exp(-\gamma t) \cos^2 \omega t \right)$$

$\omega^2 \sim \omega_0^2 = \frac{k}{m} \rightarrow m \omega^2 = k$

$$= \frac{1}{2} A_0^2 \exp(-\gamma t) k \longrightarrow E(t) = E_0 \exp(-\gamma t), \quad E_0 = \frac{1}{2} A_0^2 k$$

Time factor $\tau = 1/\gamma$

$$E(t) = E_0 \exp(-t/\tau)$$

$$\text{Also, } \frac{dE}{dt} = \frac{d}{dt} \left(\frac{1}{2} m v^2 + \frac{1}{2} k x^2 \right) = m v \frac{dv}{dt} + k x \frac{dx}{dt} = v (m a + k x) = -b v^2$$

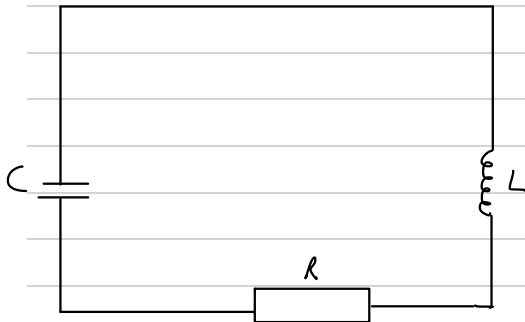
$$\text{recall } \frac{dU}{dt} = -P$$

Quality factor:

$$Q = \omega_0 / \gamma = \frac{\text{Energy stored in oscillator}}{\text{Energy dissipated per radian}} \longrightarrow x'' + \frac{\omega_0}{Q} x' + \omega_0^2 x = 0$$

$$\begin{aligned} \omega^2 &= \omega_0^2 - \frac{\gamma^2}{4} \longrightarrow \omega = \omega_0 \sqrt{1 - \frac{1}{4Q^2}} \\ &= \omega_0^2 \left(1 - \frac{1}{4Q^2} \right) \end{aligned}$$

RLC Circuits



$$0 = V_C + V_L + V_R$$

$$0 = \frac{q}{C} + \frac{dI}{dt} L + R I$$

$$0 = q'' + \frac{R}{L} q' + \frac{1}{LC} q$$

Equivalents:

$$\begin{array}{ll} m \Leftrightarrow L & b \Leftrightarrow R \\ \omega_0 \Leftrightarrow \sqrt{\frac{1}{LC}} & k \Leftrightarrow \frac{1}{L} \\ \gamma \Leftrightarrow \frac{R}{L} & \tau \Leftrightarrow \frac{L}{R} \end{array}$$

$$Q \Leftrightarrow \omega_0 / \gamma \rightarrow \sqrt{\frac{1}{LC}} \cdot \frac{L}{R} \rightarrow \frac{1}{R} \sqrt{\frac{L}{C}}$$

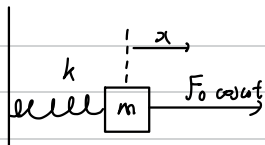
$$\omega \Leftrightarrow \omega_0^2 - \frac{\gamma^2}{4} \rightarrow \frac{1}{LC} - \frac{R^2}{4L^2}$$

3. Forced Oscillations

Undamped forced harmonic oscillator:

Equation of motion:

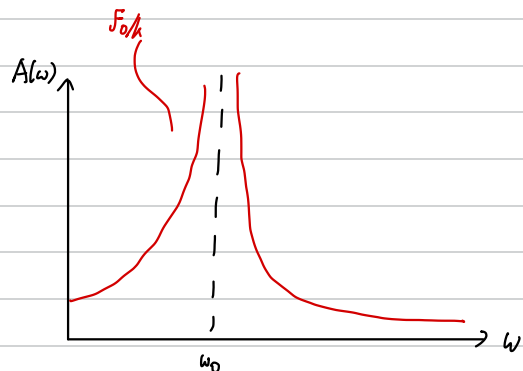
$$m\ddot{x} + kx = F_0 \cos \omega t$$



$$m\ddot{x} + kx = ka \cos \omega t, \quad F_0 = ka$$

Sol'n: $x = A(\omega) \cos(\omega t - \delta)$, $a = F_0/k$

where:
$$\begin{cases} A(\omega) = \frac{a}{1 - \omega^2/\omega_0^2}, \quad \delta = 0, \text{ for } \omega_0 > \omega \\ A(\omega) = \frac{-a}{1 - \omega^2/\omega_0^2}, \quad \delta = \pi, \text{ for } \omega_0 < \omega \end{cases}$$



Forced oscillations with damping:

Equation of motion:

$$m\ddot{x} + b\dot{x} + kx = F_0 \cos \omega t$$

$$\ddot{x} + r\dot{x} + \omega_0^2 x = F_0/m \cos \omega t, \quad r = \frac{b}{m}, \quad \omega_0^2 = \frac{k}{m}$$

Sol'n:

$$x = A(\omega) \cos(\omega t - \delta)$$

where,
$$\begin{cases} A(\omega) = \frac{a\omega_0^2}{\sqrt{(\omega_0^2 - \omega^2)^2 + \omega^2 r^2}} \\ \tan \delta = \frac{\omega r}{\omega_0^2 - \omega^2} \end{cases}$$

$0 \leq \delta \leq \pi$

Max amplitude, occurs when $\frac{d[(\omega_0^2 - \omega^2)^2 + \omega^2 r^2]}{d\omega} = 0$

$$\frac{2r^2\omega + 2(\omega_0^2 - \omega^2) \cdot 2\omega}{2\sqrt{(\omega_0^2 - \omega^2)^2 + \omega^2 r^2}} = 0$$

$$2r^2\omega - 4\omega_0^2\omega + 4\omega^3 = 0$$

$$\omega^2 = \omega_0^2 - \frac{r^2}{2}$$

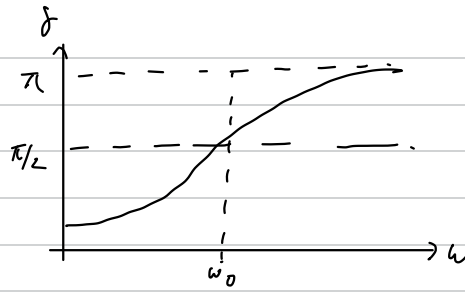
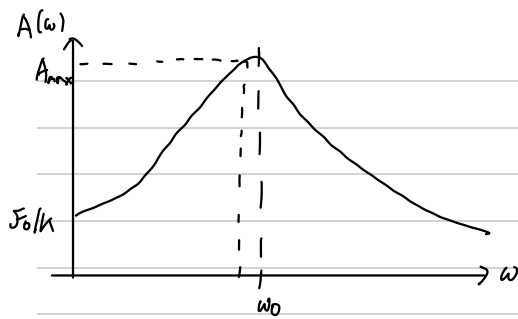
Resonance Frequency

$$\omega_{\text{res}} = \omega \sqrt{1 - \frac{r^2}{2\omega_0^2}}$$

$$\begin{aligned} A(\omega_{\text{res}}) &= \frac{a\omega_0^2}{\sqrt{[\omega_0^2 - \omega_0^2(1 - \frac{r^2}{2\omega_0^2})]^2 + \omega_0^2(1 - \frac{r^2}{2\omega_0^2})r^2}} \\ &= \frac{a\omega_0^2}{\sqrt{(\frac{r^2}{2})^2 + \omega_0^2(1 - \frac{r^2}{2\omega_0^2})r^2}} \\ &= \frac{a\omega_0^2}{\sqrt{\frac{r^4}{4} + \omega_0^2\frac{r^2}{2} - \frac{r^4}{2}}} \end{aligned}$$

$$\begin{aligned} &= \frac{a\omega_0^2}{\sqrt{\omega_0^2\frac{r^2}{4} - \frac{r^4}{4}}} \\ &= \frac{a\omega_0/r}{\sqrt{1 - \frac{r^2}{4\omega_0^2}}} \end{aligned}$$

or
$$= \frac{aQ}{\sqrt{1 - \frac{1}{4Q^2}}}$$



Velocity, power for forced, damped oscillators:

$$\text{Let } v = -v_0(\omega) \sin(\omega t - \delta), \quad v_0(\omega) = A(\omega)\omega = \frac{a\omega\omega_0^2}{\sqrt{(\omega_0^2 - \omega^2)^2 + \omega^2\gamma^2}}$$

$$= \frac{a\omega_0^2}{\sqrt{\left(\frac{\omega_0}{\omega} - \frac{\omega}{\omega_0}\right)^2 \omega_0^2 + \gamma^2}} \quad \begin{cases} \omega \rightarrow 0, v_0(\omega) \rightarrow 0 \\ \omega \rightarrow \infty, v_0(\omega) \rightarrow 0 \\ \omega \rightarrow \omega_0, v_0(\omega) \rightarrow \text{max} = a\omega_0^2/\gamma \end{cases}$$

$$\begin{aligned} P(t) &= F \cdot v \\ &= b v^2(t) \\ &= b v_0^2(\omega) \sin^2(\omega t - \delta) \end{aligned}$$

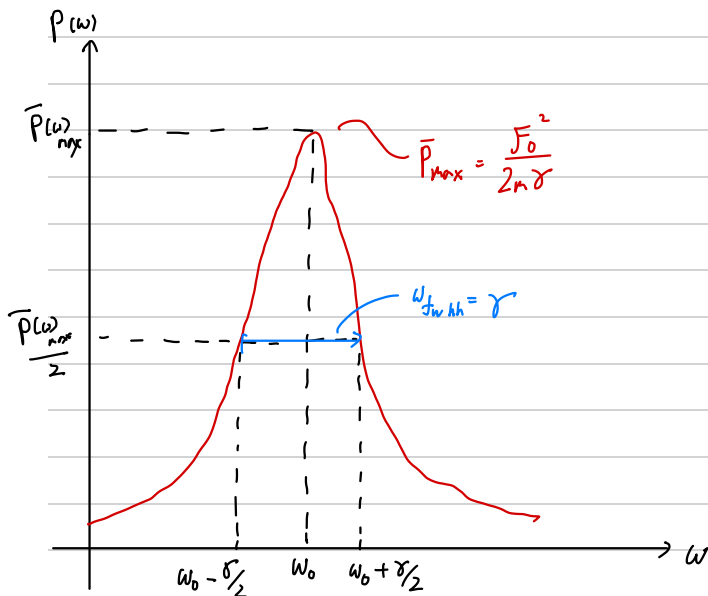
Instantaneous Power

$$\begin{aligned} \bar{P}(\omega) &= \frac{1}{T} \int_{t_0}^{t_0+T} P(t) dt \\ &= \frac{b v_0^2(\omega)}{T} \int_{t_0}^{t_0+T} \sin^2(\omega t - \delta) dt \\ &= \frac{b v_0^2(\omega)}{2} \\ &= \frac{b}{2} \left(\frac{a^2 \omega^2 \omega_0^4}{(\omega_0^2 - \omega^2)^2 + \omega^2 \gamma^2} \right) \end{aligned}$$

$$\begin{aligned} &= \frac{1}{T} \int_{t_0}^{t_0+T} \frac{1}{2} (1 - \cos 2(\omega t - \delta)) dt \\ &= \frac{1}{2} \left(t - \frac{\sin 2(\omega t - \delta)}{2\omega} \right) \Big|_{t_0}^{t_0+T} \\ &= \frac{1}{2} \left(t_0 + T - \frac{\sin 2(\omega(t_0+T) - \delta)}{2\omega} - t_0 + \frac{\sin 2(\omega t_0 - \delta)}{2\omega} \right) \\ &= \frac{1}{2} \left(t_0 + T - \frac{\sin(2\omega t_0 - \delta + 4\pi)}{2\omega} - \frac{2\omega T}{2\omega} - t_0 + \frac{\sin 2(\omega t_0 - \delta)}{2\omega} \right) \\ &= T/2 \end{aligned}$$

Average power over one period of oscillation

$$= \frac{\omega^2 F_0^2 \gamma}{2m[(\omega_0^2 - \omega^2)^2 + \omega^2 \gamma^2]}$$



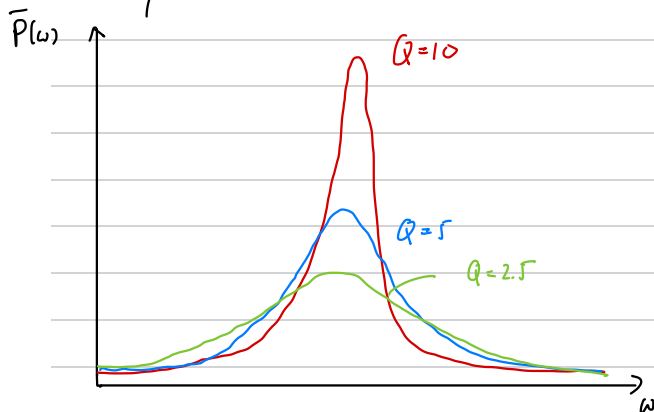
Power Resonance Curve

$$\begin{cases} \omega \rightarrow 0, \bar{P}(\omega) \rightarrow 0 \\ \omega \rightarrow \infty, \bar{P}(\omega) \rightarrow 0 \end{cases}$$

$$\begin{aligned} \text{If } \omega \approx \omega_0, \\ (\omega^2 - \omega_0^2) \approx 2\omega_0(-\Delta\omega) \end{aligned}$$

$$\text{Then } \bar{P}(\omega) = \frac{F_0^2}{2m\gamma \left(\frac{4\Delta\omega^2}{\gamma^2} + 1 \right)}$$

Quality factor and Power Curve

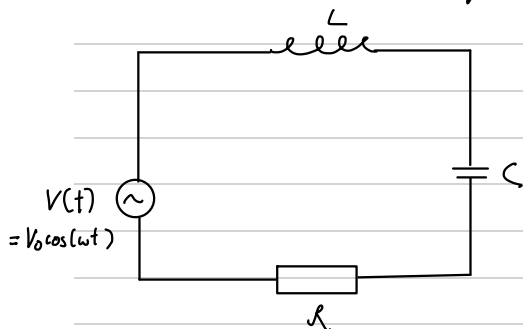


$$\text{Since } Q = \frac{\omega_0}{\gamma} = \frac{\omega_0}{\omega_{\text{width}}} = \frac{\omega_0}{2\Delta\omega}$$

$$\bar{P}(\omega) = \frac{F_0^2}{2m\omega_0 Q \left(4 \left(\frac{\Delta\omega}{\omega_0} \right)^2 + \frac{1}{Q^2} \right)}, \quad Q = \frac{\text{resonance freq.}}{\text{width}}$$

→ Higher $Q \rightarrow$ narrower, uneven distribution, high peak
 Lower $Q \rightarrow$ wider, even distribution, low peak.

RLC Circuit with AC voltage:



$$V_0 \cos(\omega t) = L \frac{d^2 q}{dt^2} + R \frac{dq}{dt} + \frac{q}{C}$$

$$\frac{V_0}{L} \cos(\omega t) = \frac{d^2 q}{dt^2} + \frac{R}{L} \frac{dq}{dt} + \frac{q}{LC}$$

Equivalent_{LC}:

$$m \Leftrightarrow L, \quad b \Leftrightarrow R$$

$$\omega_0 \Leftrightarrow \sqrt{\frac{1}{LC}}, \quad k \Leftrightarrow \frac{1}{C}$$

$$\gamma \Leftrightarrow \frac{R}{L}, \quad \tau \Leftrightarrow \frac{L}{R}$$

$$Q \Leftrightarrow \frac{1}{R} \sqrt{\frac{L}{C}}, \quad a \Leftrightarrow CV_0$$

$$\text{Then, } q_f(t) = q_0(\omega) \cos(\omega t - \delta)$$

$$q_0(\omega) = \frac{CV_0 \cdot \frac{1}{LC}}{\sqrt{\left(\frac{1}{LC} - \omega^2\right)^2 + \frac{\omega^2 R^2}{L^2}}} = \frac{V_0}{\omega \sqrt{\left(\frac{1}{\omega C} - \omega L\right)^2 + R^2}}$$

$$I = \frac{dq}{dt} = -\omega q_0(\omega) \sin(\omega t - \delta) = -I_0(\omega) \sin(\omega t - \delta), \quad I_0(\omega) = \omega q_0(\omega)$$

$$= \frac{-V_0 \sin(\omega t - \delta)}{\sqrt{\left(\frac{1}{\omega C} - \omega L\right)^2 + R^2}}$$

Voltage for each part:

$$V_R = RI$$

$$V_C = \frac{q}{C}$$

$$V_L = L \frac{dI}{dt}$$

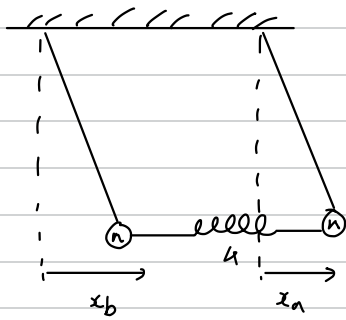
$$= \frac{-V_0 R \sin(\omega t - \delta)}{\sqrt{\left(\frac{1}{\omega C} - \omega L\right)^2 + R^2}}$$

$$= \frac{V_0 \cos(\omega t - \delta)}{\omega C \sqrt{\left(\frac{1}{\omega C} - \omega L\right)^2 + R^2}}$$

$$= \frac{-V_0 \omega L \cos(\omega t - \delta)}{\sqrt{\left(\frac{1}{\omega C} - \omega L\right)^2 + R^2}}$$

4 Coupled oscillators

Normal modes of oscillation



Equations of motion:

$$m_a x_a'' = -k(x_a - x_b) - \frac{mg}{L} x_a \longrightarrow x_a'' = -\left(\frac{g}{L} + \frac{k}{m}\right)x_a + \frac{k}{m}x_b$$

$$m_b x_b'' = -k(x_b - x_a) - \frac{mg}{L} x_b \longrightarrow x_b'' = \frac{k}{m}x_a - \left(\frac{g}{L} + \frac{k}{m}\right)x_b$$

Let $x_a = A \cos \omega t$, $x_b = B \cos \omega t$
 $x_a'' = -\omega^2 A \cos \omega t$, $x_b'' = -\omega^2 B \cos \omega t$

$$\longrightarrow -\omega^2 \begin{bmatrix} A \\ B \end{bmatrix} = \begin{bmatrix} -\left(\frac{g}{L} + \frac{k}{m}\right) & \frac{k}{m} \\ \frac{k}{m} & -\left(\frac{g}{L} + \frac{k}{m}\right) \end{bmatrix} \begin{bmatrix} A \\ B \end{bmatrix}$$

$$\longrightarrow \begin{vmatrix} -\left(\frac{g}{L} + \frac{k}{m}\right) + \omega^2 & \frac{k}{m} \\ \frac{k}{m} & -\left(\frac{g}{L} + \frac{k}{m}\right) + \omega^2 \end{vmatrix} = 0$$

$$\longrightarrow \omega^4 - 2\omega^2\left(\frac{g}{L} + \frac{k}{m}\right) + \left(\frac{g}{L} + \frac{k}{m}\right)^2 - \left(\frac{k}{m}\right)^2 = 0$$

$$\omega^4 - 2\omega^2\left(\frac{g}{L} + \frac{k}{m}\right) + \left(\frac{g}{L}\right)^2 + \frac{2gk}{Ln} = 0$$

$$\omega^4 - 2\omega^2\left(\frac{g}{L} + \frac{k}{m}\right) + \left(\frac{g}{L}\right)\left(\frac{g}{L} + \frac{2k}{n}\right) = 0$$

$$\left(\omega^2 - \frac{g}{L}\right)\left(\omega^2 - \left(\frac{g}{L} + \frac{2k}{n}\right)\right) = 0$$

$$\omega_1 = \sqrt{\frac{g}{L}}, \quad \omega_2 = \sqrt{\frac{g}{L} + \frac{2k}{n}}$$

And to find the nature of normal modes:

For ω_1 :

$$\begin{bmatrix} -\left(\frac{g}{L} + \frac{k}{m}\right) + \frac{g}{L} & \frac{k}{m} \\ \frac{k}{m} & -\left(\frac{g}{L} + \frac{k}{m}\right) + \frac{g}{L} \end{bmatrix} \vec{v} = \vec{0}$$

$$\begin{bmatrix} -\frac{k}{m} & \frac{k}{m} \\ \frac{k}{m} & -\frac{k}{m} \end{bmatrix} \vec{v} = \vec{0}$$

$$\vec{v} = \begin{bmatrix} A \\ B \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

For ω_2 :

$$\begin{bmatrix} -\left(\frac{g}{L} + \frac{k}{m}\right) + \frac{g}{L} + \frac{2k}{n} & \frac{k}{m} \\ \frac{k}{m} & -\left(\frac{g}{L} + \frac{k}{m}\right) + \frac{g}{L} + \frac{2k}{n} \end{bmatrix} \vec{v} = \vec{0}$$

$$\begin{bmatrix} \frac{k}{m} & \frac{k}{m} \\ \frac{k}{m} & \frac{k}{m} \end{bmatrix} \vec{v} = \vec{0}$$

$$\vec{v} = \begin{bmatrix} A \\ B \end{bmatrix} = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

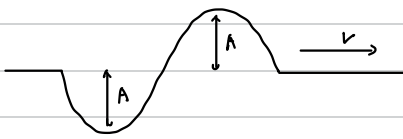
Equation of General Motion:

$$x_a = \frac{1}{2} \left[C_1 \cos(\omega_1 t + \phi_1) + C_2 \cos(\omega_2 t + \phi_2) \right]$$

$$x_b = \frac{1}{2} \left[C_1 \cos(\omega_1 t + \phi_1) - C_2 \cos(\omega_2 t + \phi_2) \right]$$

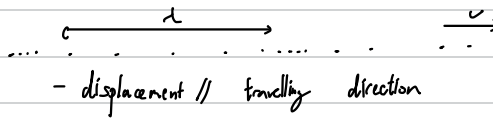
5. Travelling waves

Transverse waves:



- Displacement $\perp$ to travelling direction

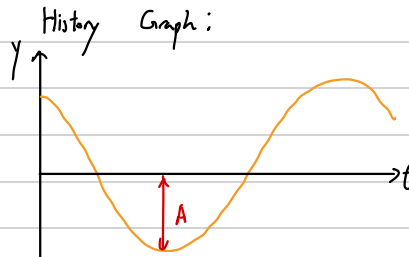
Longitudinal waves:



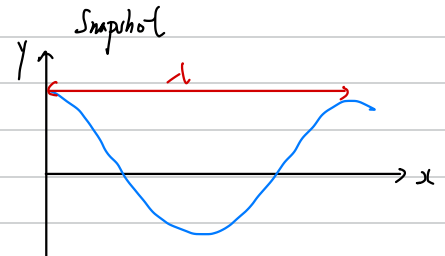
- displacement $\parallel$ travelling direction

Sinusoidal waves:

$$y(x, t) = y_m \sin(kx + \omega t + \phi_0)$$



History Graph:



Snapshot

where:

- Angular wave number, $k = 2\pi/\lambda$ [rad m^{-1}] - Fixed point, t varies

- Fixed time, x varies

- Angular frequency, $\omega = 2\pi/T$ [rad s^{-1}]

- Phase constant, ϕ_0 , determines $y @ x=0, t=0$

- y_m , amplitude

- v_e , velocity = $f\lambda = \omega/k$

1D. Wave equation:

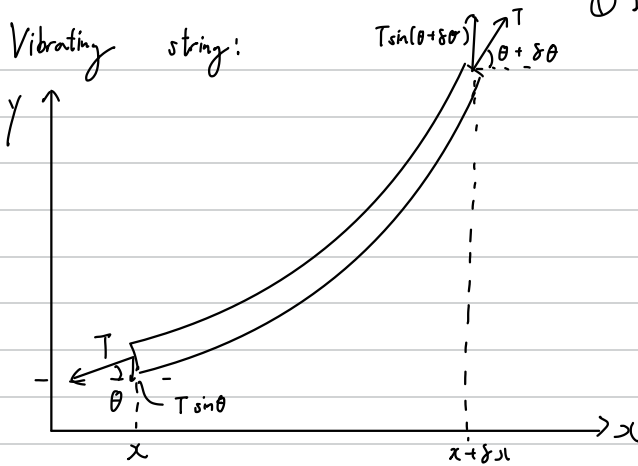
$$\frac{\partial^2 y}{\partial t^2} = v_e^2 \frac{\partial^2 y}{\partial x^2}$$

where $y(x, t) = f(x - vt) + g(x + vt)$

← +ve direction

← -ve direction

Vibrating string:



① In y-direction:

Apply small angle approximation:
 $\theta \sim \sin \theta \sim \tan \theta \sim \frac{\partial y}{\partial x}$

$F_y = T \frac{\partial y}{\partial x}$, Force of string at any point

$$\left. \frac{\partial y}{\partial x} \right|_{x+\delta x} = \left. \frac{\partial y}{\partial x} \right|_x + \frac{\partial}{\partial x} \left(\frac{\partial y}{\partial x} \right) \delta x, \text{ linear approximation}$$

$$= \left(\frac{\partial y}{\partial x} \right)_x + \left(\frac{\partial^2 y}{\partial x^2} \right) \delta x$$

With:

- Uniform density μ
- Length L

$$F_y|_{x+\delta x} = T \left(\left. \frac{\partial y}{\partial x} \right|_x + \left(\frac{\partial^2 y}{\partial x^2} \right) \delta x \right)$$

Resultant force (Force @ $x+\delta x$ - Force @ x):

$$T \left(\frac{\partial^2 y}{\partial x^2} \right) \delta x$$

② In x-direction:

Resultant force:

$$T \cos(\theta + \delta \theta) - T \cos \theta \sim 0 \quad \text{for } \cos(\theta + \delta \theta) \sim \cos \theta$$

③ Apply $F = ma$

$$T \left(\frac{\partial^2 y}{\partial x^2} \right) \delta x = \mu \delta x \left(\frac{\partial^2 y}{\partial t^2} \right)$$

$$\frac{\partial^2 y}{\partial t^2} = \frac{T}{\mu} \left(\frac{\partial^2 y}{\partial x^2} \right), \quad v = \sqrt{\frac{T}{\mu}}$$

Energy of vibrating string

$$K = \frac{1}{2} \mu \delta x \left(\frac{\partial y}{\partial t} \right)^2$$

U (String potential energy, extension $\times$ tension):

$$= T(\delta s - \delta x)$$

Small angle

Taylor expansion

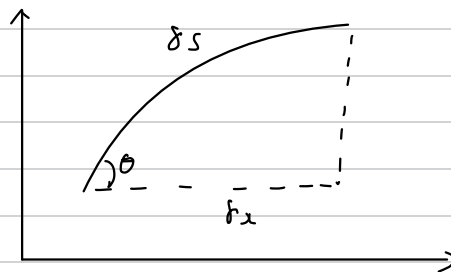
$$\delta s = \frac{\delta x}{\cos \theta} = \frac{\delta x}{(1 - \sin^2 \theta)^{1/2}} \approx \frac{\delta x}{(1 - \theta^2)^{1/2}} \approx \delta x \left(1 + \frac{1}{2} \theta^2 \right) \approx \delta x \left(1 + \frac{1}{2} \left(\frac{\partial y}{\partial x} \right)^2 \right)$$

$$= \frac{1}{2} T \delta x \left(\frac{\partial y}{\partial x} \right)^2$$

$$dE = \frac{1}{2} \mu \delta x \left(\frac{\partial y}{\partial t} \right)^2 + \frac{1}{2} T \delta x \left(\frac{\partial y}{\partial x} \right)^2$$

$$\hookrightarrow E = \frac{1}{2} \mu \omega^2 A^2 l, \text{ integrated over wave length}$$

$$K_{\text{total}} = U_{\text{total}} = \frac{1}{2} \mu \omega^2 A^2 l$$



Power, P : $A^2 \mu v \omega^2 \sin^2(kx - \omega t + \phi_0) = -\tau \frac{\partial y}{\partial x} \frac{\partial y}{\partial t}$

Average Power, $\bar{P}$: $\frac{1}{2} \mu \omega^2 A^2 v = \frac{1}{2} \mu \omega^2 A^2 \frac{1}{k}$

Max Power: $A^2 \mu v \omega^2 = \sqrt{\mu \tau} A^2 \omega^2$

Wave at discontinuity:

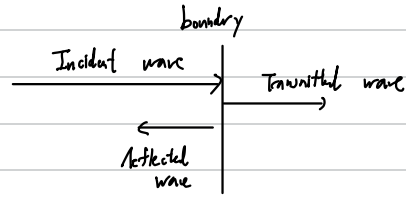
$y_i = A_i \cos(\omega t - k_1 x)$

$y_T = A_T \cos(\omega t - k_2 x)$

$y_R = A_R \cos(\omega t + k_1 x)$

$y_i + y_R = y_T$

$A_i \cos(\omega t - k_1 x) + A_R \cos(\omega t + k_1 x) = A_T \cos(\omega t - k_2 x)$



Conditions:

1. Displacements are cont. at boundary $\rightarrow A_i + A_R = A_T \rightarrow A_i + R A_i = T A_i$

2. $\frac{\partial y}{\partial x}$ is cont. at boundary $\rightarrow A_i Z_1 - A_R Z_1 = A_T Z_2 \rightarrow A_i Z_1 - R A_i Z_1 = T A_i Z_2$

$Z_1 = \frac{\tau}{v_1} = \frac{k_1}{\omega}$

Transmission coefficients:

Reflection coeff, $R = \frac{A_R}{A_i} = \frac{Z_1 - Z_2}{Z_1 + Z_2} \in [-1, 1]$

$\hookrightarrow Z_2 > Z_1$, -ve, reflected and incident wave out of phase, phase difference of π

$\hookrightarrow Z_2 < Z_1$, +ve, reflected and incident wave in phase

Transmission coeff, $T = \frac{A_T}{A_i} = \frac{2 Z_1}{Z_1 + Z_2}$, always +ve

$\hookrightarrow$ Transmitted wave, Incident wave always in phase

$1 + R = T$

Special cases:

Rigid wall, $Z_2 \rightarrow \infty$

$T = 0, R = -1$

$\hookrightarrow$ Nothing transmitted, wave completely reflected with phase change π

Free boundary, $Z_2 \rightarrow 0$

$R = 1$

$T = 2$

$\hookrightarrow$ Wave reflects completely in phase, transmitted wave has double amplitude.

Power coefficients:

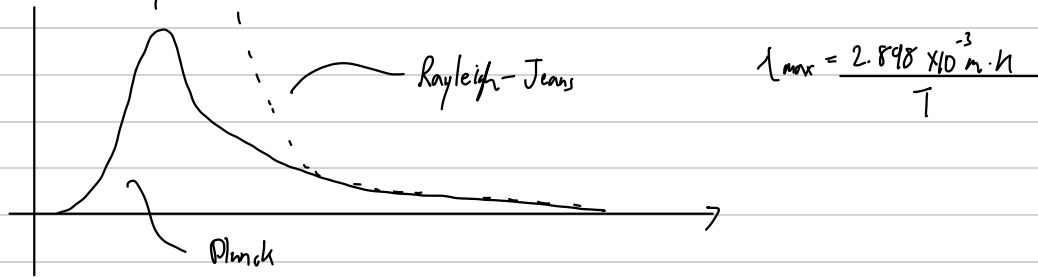
$R_c \equiv R^2 = \frac{A_R^2}{A_i^2} = \left(\frac{Z_1 - Z_2}{Z_1 + Z_2} \right)^2$, $T_c \equiv \frac{Z_2}{Z_1} T^2 = \frac{Z_2}{Z_1} \left(\frac{A_T}{A_i} \right)^2 = \frac{4 Z_2 Z_1}{(Z_1 + Z_2)^2}$, $R_c + T_c = 1$

$\hookrightarrow \frac{\text{Reflected } E}{\text{Incident } E}$

$\hookrightarrow \frac{\text{Transmitted } E}{\text{Incident } E}$

Modern Physics

4.2 Blackbody radiation



↳ Rayleigh-Jeans:

$$M_{\lambda}(\lambda, T) = \frac{2\pi c k_B T}{\lambda^4} \rightarrow \text{accurate at long } \lambda, \text{ diverges at short } \lambda, \text{ ultra violet catastrophe}$$

↳ Planck:

- Discrete, $E_n = n h f$, $n = 0, 1, 2$, discrete energy levels
- Each $h f \rightarrow$ corresponds to 1 photon,
- $h = 6.626 \times 10^{-34} \text{ J.s}$

$$M_{\lambda}(\lambda, T) = \frac{2\pi h c^2}{\lambda^5} \left(\frac{1}{e^{h c / (\lambda k_B T)} - 1} \right)$$

Taylor expand:

$$\begin{aligned} M_{\lambda} &= \frac{2\pi h c^2}{\lambda^5} \left(\frac{1}{1 + \frac{h c}{\lambda k_B T} + \frac{1}{2} \left(\frac{h c}{\lambda k_B T} \right)^2 + \dots - 1} \right) \\ &= \frac{2\pi h c^2}{\lambda^5} \left(\frac{\lambda k_B T}{h c} \right) \quad 0 \text{ as } \lambda \rightarrow \infty \\ &= \frac{2\pi c k_B T}{\lambda^4} \end{aligned}$$

4.5 Photoelectric Effect

For stopping potential V_s , maximum KE of electron:

$$e V_s = K_{\max}$$

Classically:

$E \propto \text{intensity}$

Any λ can eject electrons

Stronger light $\rightarrow$ more E^-

Experimentally:

KE indep. of intensity

No e^- ejected below f_0 , threshold frequency

intensity affects number of emitted e^-

Side note, $P = I A$,
 $I \rightarrow \text{intensity}$
 $A \rightarrow \text{area}$

Einstein's explanation

- Light consists of photons, with energy $E = h f$
- Photon is massless, $E = p c = h c / \lambda$, where $p = h / \lambda$
- 1 photon transfer E to 1 electron.

$$h f = \phi + K_{\max}$$

metals work function

$$\phi = f_0 h$$

- Greater intensity, more photons, not more energy / photon



4.4 X-ray and Bragg Diffraction:

X-rays are:

- EM wave with 0.001 - 1 nm wavelength
- High photo energy $\geq 100 \text{ eV}$

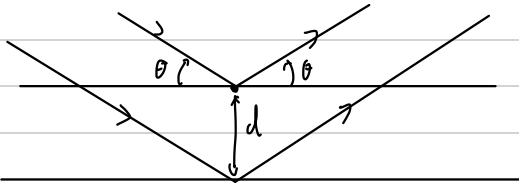
Bremsstrahlung:

e^- slow down when deflected by nuclei, cont. X-ray spectrum

$$E_r = E_{\text{outer}} - E_{\text{inner}}$$

Bragg Diffraction:

Constructive interference: $2d \sin \theta = n\lambda$



4.5 X-ray Spectra

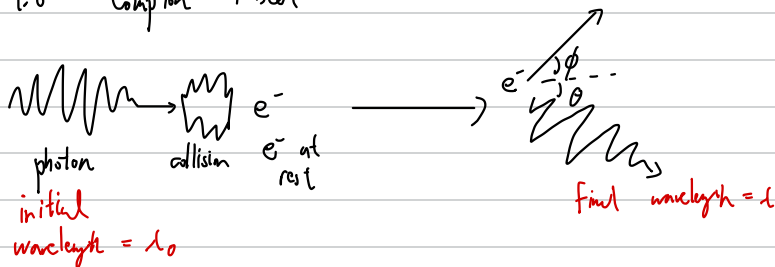
Characteristic X-ray:

Incoming e^- knocks an orbital e^- from an inner shell out of atom, an outer shell e^- fills the hole, emitting discrete energy:

Maximum freq.

$$hf_{\text{max}} = V_0 e$$

4.6 Compton Effect



$$-1 \leq \cos \theta \leq 1 \quad \text{for } 0 \leq \theta \leq \pi$$

$$\Delta \lambda = \lambda - \lambda_0 = \frac{h}{mc} (1 - \cos \theta)$$

Theory 5.2 Atomic spectrum

- Specific wavelength absorbed by elements $\rightarrow$ absorption spectrum
- If same gas is heated, it emit lights of equal wave $\rightarrow$ emission spectrum



* 5.3 Balmer - Lydberg Formula

$$\frac{1}{\lambda} = R \left(\frac{1}{n'^2} - \frac{1}{n^2} \right) \longrightarrow E = hcR \left(\frac{1}{n'^2} - \frac{1}{n^2} \right)$$

where $n > n'$,

Theory 5.4 Problem of atomic stability

$\hookrightarrow$ instability of e^- radiating energy, spiral into nucleus

Theory 5.5 Bohr's Explanation of Atomic Spectra

Postulates:

1. e^- move in stationary orbits, not radiating energy
2. Quantized energy
3. Photon emission occurs when e^- jumps b/w energy levels
 $E_r = E_i - E_{n'}$

Spectral lines:

$$\lambda = \frac{hc}{E_r}$$

* 5.6 Bohr Model of Hydrogen Atom

Classical Mechanics

1.) Centripetal Force = Coulomb Force

$$\frac{mv^2}{r} = \frac{ke^2}{r^2} \Rightarrow mv^2 = \frac{ke^2}{r}$$

$$2) KE = \frac{1}{2}mv^2 = \frac{ke^2}{2r}, U = \frac{-ke^2}{r} \text{ (in proton field)}$$

$$3) E = -K + U = -\frac{ke^2}{2r}$$

Bohr's Quantization of angular momentum

$$L = mvr = n\hbar, \hbar = \frac{h}{2\pi} \implies v = \frac{n\hbar}{mr}$$

Bohr radius:

$$m \frac{n^2 \hbar^2}{n^2 r^2} = \frac{ke^2}{r}, \text{ since } mv^2 = \frac{ke^2}{r}$$

$$r = \frac{n^2 \hbar^2}{mke^2}$$

$$= a_0 n^2 \sim a_0, \text{ Bohr radius} = \frac{4\pi \epsilon_0 \hbar^2}{e^2 m} \text{ since } h = \frac{1}{4\pi \epsilon_0}$$

Quantized Energy levels

$$E = \frac{-ke^2}{2r} = \frac{-ke^2}{2a_B n^2}$$

Rydberg Energy, $E_R = \frac{ke^2}{2a_B} = 13.6 \text{ eV}$

Allowed energies: $\frac{-13.6 \text{ eV}}{n^2}$

Connection to Rydberg's formula: $E_R \left(\frac{1}{n_f^2} - \frac{1}{n_i^2} \right)$

★ 5.8 Hydrogen-Like Ions

↳ Applicable for atoms that has a single e^- in orbit, with nucleus charge of $+Ze$

Replace $\frac{ke^2}{r^2}$ by $\frac{Zke^2}{r^2}$

$$\frac{mv^2}{r} = \frac{Zke^2}{r^2}, \quad mvr = n\hbar$$
$$v = \frac{n\hbar}{mr}$$

$$\frac{m \frac{n^2 \hbar^2}{r^2}}{m} = \frac{Zke^2}{r}$$
$$r = \frac{n^2 \hbar^2}{Zke^2 m} = n^2 \frac{a_B}{Z}$$

Energy: $KE = \frac{1}{2} mv^2 = \frac{1}{2} \frac{Zke^2}{r}$

$$U = -\frac{Zke^2}{r}$$

$$E = -\frac{1}{2} \frac{Zke^2}{r}$$
$$= -\frac{1}{2} Zke^2 \cdot \frac{Z}{n^2 a_B}$$

$$= -\left(\frac{ke^2}{2a_B} \right) \frac{Z^2}{n^2}$$

$$= -E_R \left(\frac{Z^2}{n^2} \right)$$

★ 5.9 X-ray Spectra

↳ $E_n = \frac{-Z^2}{n} E_R$ and $E_n = hf$, suggests that

↳ $hf \propto Z$, but experimentally,
it is found that $hf \propto Z-1$ (or $E_f \propto (Z-1)^2$)

↳ From assumption that inner e^- only feels force of nucleus

↳ But in reality, some electrons "screen" it $\rightarrow$ small reduction in nuclear charge, represented by $(Z-\sigma)$

Theory 5.10 Other evidence for Atomic energy levels

Frank-Hertz experiment

↳ e^- accelerated through Hg vapor

↳ Current drops every 4.9V, e^- lose fixed amounts of Energy

Energy-Loss Spectra (He gas)

↳ Inelastic scattering of e^- shows discrete energy losses

* 6.2 De Broglie's Hypothesis

- Idea that all matter has wave-like properties, idea of **matter waves**

$$\lambda = \frac{h}{p} = \frac{h}{mv}$$

Wave nature to quantized angular momentum:

↳ e^- wave must fit around orbit's circumference

$$\hookrightarrow 2\pi r = n\lambda$$

$$2\pi r = \frac{n h}{p}$$

$$L = pr = n\hbar$$

Theory 6.3 Experimental verification

Davisson & Germer firing e^- at nickel,

↳ Observed diffraction pattern

↳ Wavelength matches de Broglie prediction

G.P. Thomson demonstrated e^- diffraction through thin metal foils

6.4 Quantum Wave Function

Classical wave:

$$\hookrightarrow y(x,t) = A \sin(kx - \omega t)$$

$$= \text{Re} [A e^{i(kx - \omega t)}]$$

$$\hookrightarrow \text{where } A e^{i(kx - \omega t)} = A [\cos(kx - \omega t) + i \sin(kx - \omega t)]$$

$$\Phi = \Phi_{\text{real}} + i \Phi_{\text{imaginary}}$$

$$|\Phi| = \sqrt{\Phi_{\text{real}}^2 + \Phi_{\text{imaginary}}^2} = A$$

in classical waves, $\Phi_{\text{imaginary}}$ is not physically meaningful

Matter wave:

$$\Phi(x,t) = A e^{i(kx - \omega t)}$$

↳ Mathematical description of a system

↳ Encodes all measurable information

Max Born's interpretation:

↳ Ψ is a probability amplitude, not a physical medium wave.

↳ $|\Psi|^2 = \Psi_{\text{real}}^2 + \Psi_{\text{imag}}^2$, probability density of finding the particle at r , at time t .

↳ Ψ may be complex, but $|\Psi|^2$ is real, positive, measurable

Amplitude $\rightarrow |\Psi|$; Probability density $\rightarrow |\Psi|^2$

Phase $\rightarrow \phi$, governs momentum, interference, time evolution

Amplitude-phase form:

$$\Psi(x,t) = R(x,t) e^{i\phi(x,t)}$$

Properties of wave function:

↳ Single valued, continuous, finite

Ψ must have only 1 value at each point, smooth change without jumps, stay finite

↳ Continuous first derivative

$\frac{d\Psi}{dx}$ is cont. (except at infinite potential steps) $\rightarrow$ Physical quantities are well defined

↳ Normalizable:

$$\int_{-\infty}^{\infty} |\Psi(x)|^2 dx = 1$$

6.5 Which slit does e^- go through?

Double Slit experiment:

- Interference disappears after observation.

↳ Measurement disturbs the wave

$$\lambda_d \approx d$$

$$p_{el} = \frac{h}{\lambda_{el}} \approx \frac{h}{d}$$

But measuring using light, the beam has to be narrower than d ,

$$\lambda_r \leq d$$

$$\text{then } p_r = \frac{h}{\lambda_r} \geq \frac{h}{d} \rightarrow \text{alters path of electron}$$

6.6 Sinusoidal waves

$$y(x,t) = A \sin(kx + \omega t + \phi)$$

$$= A e^{i(kx + \omega t + \phi)}$$

Angular freq = $\omega = 2\pi f = \frac{2\pi}{T}$

Phase

wavenumber $\frac{2\pi}{\lambda}$

Amplitude

$$E = hf = \hbar \cdot 2\pi f = \hbar \omega$$

$$p = \frac{h}{\lambda} = \hbar \frac{2\pi}{\lambda} = \hbar k$$

$$\text{Thus, } \Psi(x,t) = A e^{i(kx + Et/\hbar + \phi)}$$

6.7 Wave packets, Fourier Analysis, Also 6.8, 6.9 Uncertainty Principles

- ↳ Pure sinusoidal has a well-defined wavenumber, momentum, but not position (infinite)
- ↳ A particle/wave must exist only within a finite region

Fourier Analysis

- ↳ Constructing localized wave (wave packets) by combining sinusoidal waves
- ↳ Superposing multiple sinusoids

Fourier Series:

$$f(x) = \sum_{n=0}^{\infty} [A_n \cos(k_n x) + B_n \sin(k_n x)]$$

Even functions, $f(-x) = f(x)$:

Odd functions use sine

use cosine terms, $k_n = \frac{2\pi n}{\lambda}$

Optional, for even functions, $A_n = \frac{2}{\pi n} \sin\left(\frac{\pi n a}{\lambda}\right)$

Fourier Integral:

as $\lambda \rightarrow \infty$, $\Delta k = \frac{2\pi}{\lambda} \rightarrow 0$ (as repeat length, $\lambda \rightarrow \infty$, the period $\rightarrow 0$, i.e. non-repeating)

$$f(x) = \int A(k) e^{ikx} dk$$

↳ $A(k) = \frac{1}{2\pi} \int f(x) e^{-ikx} dx$, Fourier transform, shows how k contribute to the wave packet.

↳ $f(x)$ describes how wave behaves in position space (localized in x)

↳ $A(k)$ gives spectrum in wavenumber space (spread in k)

Narrow $x \Rightarrow$ Broad k , Broad $x \Rightarrow$ Narrow k

Fourier Space to Hilbert Space

$$\psi(x) = \int A(k) e^{ikx} dk$$

↳ e^{ikx} basis vector

↳ $A(k)$, components of the state

↳ Inner product, $\int \phi(x) \psi^*(x) dx$ defines overlap

↳ Normalization, $\int \psi(x) \psi^*(x) dx = 1$

Fourier uncertainty relation:

$$\Delta x = \sqrt{(x - x_0)^2}, \Delta k = \sqrt{(k - k_0)^2}, k = \frac{2\pi}{\lambda}$$

$$\text{and } \Delta x \Delta k \geq \frac{1}{2}$$

Replacing fixed t , function of x to $\Rightarrow$ fixed x , function of t , gives: $\Delta t \Delta \omega \geq \frac{1}{2}$

$x \rightarrow t$

$k \rightarrow \omega$ $\frac{2\pi}{\lambda} \rightarrow \frac{2\pi}{T}$

Heisenberg's Uncertainty Principle

$$\Delta x \Delta k \geq \frac{1}{2}$$

$$\Delta x \Delta p \geq \frac{\hbar}{2}, \text{ since } p = \hbar k$$

and

$$\Delta t \Delta \omega \geq \frac{1}{2}$$

$$\Delta t \Delta E \geq \frac{\hbar}{2}, \text{ since } E = \hbar \omega$$

6.10 Velocity of Wave Packets ...

7.2, 7.3 Classical and Quantum standing waves

Classically: $\Psi(x, t) = A \sin kx \cos \omega t$, $\lambda = \frac{2a}{n}$, $\lambda = \frac{2\pi}{k} \rightarrow \frac{2\pi}{k} = \frac{2a}{n} \rightarrow k = \frac{n\pi}{a}$

$$= \psi(x) \cos \omega t$$

↑ spatial ↑ temporal

More generally, $\psi(x) [a \cos \omega t + b \sin \omega t]$

Quantum: $\cos \omega t - i \sin \omega t = e^{-i\omega t}$

$$\Psi(x) = \psi(x) e^{-i\omega t}$$

$$\rightarrow |\Psi(x, t)|^2 = |\psi(x)|^2$$

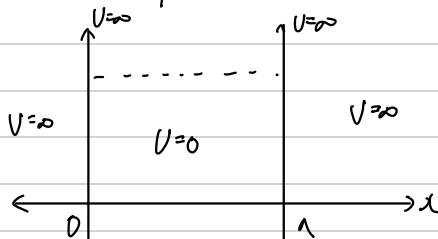
↳ Each stationary state has definite $E = \hbar \omega$

↳ Superposition of stationary states $\rightarrow$ time-dependent probability density

↳ Superposition of spatial wavefunctions $\rightarrow$ particles simultaneously occupying multiple spatial states within a potential

7.4 Particle in a rigid box

Infinite square well:



Solution:

$$\Psi(x) = A \sin kx + B \cos(kx)$$

Boundary cond:

$$\textcircled{1} \Psi(0) = 0 = B$$

$$\textcircled{2} \sin ka = 0 \quad \begin{cases} k = \frac{n\pi}{a} \\ ka = n\pi \\ \lambda = \frac{2a}{n} = \frac{2\pi}{k} \end{cases}$$

It must satisfy $\Psi(0) = \Psi(a) = 0$

↳ Allowed wave lengths, energy levels:

$$\lambda = \frac{2a}{n}, \quad E_n = \frac{p^2}{2m} \quad \text{and} \quad p = \frac{h}{\lambda} = \frac{2\pi\hbar}{\lambda} = \frac{n\pi\hbar}{a}$$

$$E_n = \frac{n^2 \pi^2 \hbar^2}{2ma^2}$$

↳ Complete wave function:

$$\Psi(x, t) = A \sin(kx) e^{-i\omega t}$$

$$= \frac{A}{2i} (e^{i(kx - \omega t)} - e^{-i(kx + \omega t)}) \quad \text{since } \sin(kx) = \frac{e^{ikx} - e^{-ikx}}{2i}$$

↑ wave moving to right wave moving to left

↳ Superposition of 2 travelling waves

7.5 The independent Schrödinger Equation

Schrodinger equation:

$$\frac{d^2 \psi(x)}{dx^2} = -k^2 \psi(x)$$

↳ Defines allowed energies E , corresponding wavefunctions $\psi(x)$
↳ Only certain E values produce valid sol $\rightarrow$ quantization of energy
↳ $\psi(x) \rightarrow$ probability of finding particle at position x

$$\text{Since } KE = \frac{p^2}{2m} = \frac{\hbar^2 k^2}{2m} \rightarrow k^2 = \frac{2mE}{\hbar^2}$$

$$\text{and } E = K + U(x)$$

Then,

$$\frac{d^2 \psi(x)}{dx^2} = \frac{2m}{\hbar^2} (U(x) - E) \psi(x)$$

7.6 Operators, observables and expectation values

Hamiltonian, eigenfunction, eigenvalue

$$\text{Hamiltonian: } \hat{H} = -\frac{\hbar^2}{2m} \frac{d^2}{dx^2} + U(x)$$

$$\text{where } \hat{H} \psi(x) = E \psi(x)$$

only certain E makes $\psi(x)$ valid.

$$\text{Operators: } \hat{x}, \hat{p}, \hat{T}_{(KE)}, \hat{U}, \hat{H}$$

Expectation value

$$\langle \hat{A} \rangle = \int \psi^* \hat{A} \psi dx, \text{ average result you would obtain by measuring many identical systems}$$

If $\psi(x)$ is an eigenfunction of $\hat{A}$, then $\langle \hat{A} \rangle = \text{eigenvalue}$.

↳ If 2 operators don't have common eigenfunctions, their corresponding quantities cannot be simultaneously determined
↳ Eigenfunctions can determine exact eigenvalues

Time-independent Schrödinger Equation

$$\frac{d^2 \psi}{dx^2} = -\frac{2mE}{\hbar^2} \psi \quad \text{Boundary cond. } \psi(0)=0, \psi(a)=0$$

~ 2 indep. solutions

$$\text{if } E < 0, -\frac{2mE}{\hbar^2} > 0 \quad \text{for } \alpha^2 = \frac{2mE}{\hbar^2}$$

$$\text{then } \psi(x) = Ae^{\alpha x} + Be^{-\alpha x}$$

$A=B=0$ for boundary cond.

$$\text{if } E > 0,$$

$$\psi(x) = A \sin(kx) + B \cos(kx), \text{ from boundary cond.}$$

$$\psi(x) = A \sin(kx), \quad ka = n\pi$$

$$E = \frac{\hbar^2 n^2 \pi^2}{2m a^2}, \quad A = \sqrt{\frac{2}{a}} \text{ from normalizing}$$

7.1 Time - Dependent Schrödinger Equation

$$i\hbar \frac{\partial \Psi(x,t)}{\partial t} = \left[-\frac{\hbar^2}{2m} \frac{\partial^2}{\partial x^2} + V(x) \right] \Psi(x,t) \quad \text{for } \Psi(x,t) = \psi(x) e^{-iEt/\hbar} / E = \hbar\omega = \hbar\omega$$
$$= \psi(x) e^{-\frac{iE t}{\hbar}}$$

Superposition:

$$\hookrightarrow \Psi(x,t) = c_1 \psi_1 e^{-iE_1 t/\hbar} + c_2 \psi_2 e^{-iE_2 t/\hbar}$$

Probability density:

$$|\Psi(x,t)|^2 = |c_1|^2 |\psi_1|^2 + |c_2|^2 |\psi_2|^2 + 2 \underbrace{c_1^* c_2}_{\text{spatial overlap}} \underbrace{\psi_1^* \psi_2}_{\text{temporal beating}} e^{-i(E_1 - E_2)t/\hbar}$$

7.0? Superposition

Time - indep.

Solving Schrödinger's Equation:

$$\hat{H} \psi_n = E_n \psi_n, \text{ produces } \psi_n \rightarrow \text{eigenfunctions}, E_n \rightarrow \text{eigenvalue } E_n$$

Satisfying:

$$\begin{aligned} \text{i.) } \int \psi_n^* \psi_m dx &= 0 \quad \text{for } n \neq m \\ \text{ii.) } \int \psi_n^2(x) dx &= 1 \end{aligned} \quad \left. \begin{array}{l} \text{in dependence} \\ \text{physical meaning} \end{array} \right\} \rightarrow \text{Orthogonal condition } \int \psi_n^*(x) \psi_m(x) dx = \delta_{nm} = \begin{cases} 1, & n=m \\ 0, & n \neq m \end{cases}$$

Completeness

It any physically valid wave function can be expressed as a linear combination of ψ_n

$$\psi(x) = \sum_n c_n \psi_n(x)$$

→ basis vectors, spans space

Finding coefficients

$$c_n = \int \psi_n^*(x) \psi(x) dx \quad \begin{array}{l} c_n \rightarrow \text{probability amplitude} \\ |c_n|^2 \rightarrow \text{probability of measuring energy } E_n \end{array}$$

$$\sum_n |c_n|^2 = 1, \text{ normalization}$$

Probability Density

$$\sum_{m,n} c_m^* c_n \psi_m^*(x) \psi_n(x)$$

Time - dep.

$$\Psi(x,t) = \sum_n c_n \psi_n(x) e^{-iE_n t/\hbar}$$

$$\sum_n |c_n|^2 = 1 //$$

7.7 The Free Particle

$$V(x) = 0 \quad \text{and} \quad -\frac{\hbar^2}{2m} \frac{d^2 \psi(x)}{dx^2} = E \psi(x)$$

↳ Energy becomes continuous

$E < 0 \rightarrow$ impossible

$E > 0$: $\psi(x) = A \sin(kx) + B \cos(kx)$ (typically for confined system)

$$\psi(x) = C e^{ikx} + D e^{-ikx}$$

left with momentum $-\hbar k$
right with momentum $+\hbar k$

$$i\hbar \frac{\partial \Psi(x,t)}{\partial t} = -\frac{\hbar^2}{2m} \frac{\partial^2 \Psi(x,t)}{\partial x^2} \quad \text{for } V(x)=0$$

$$\hookrightarrow \frac{\partial^2 \psi}{\partial x^2} = -\frac{2mE}{\hbar^2} \psi, \quad i\hbar \frac{1}{T} \frac{dT}{dt} = E \rightarrow T(t) = e^{-iEt/\hbar}$$

$$\text{Then } \Psi(x,t) = C e^{i(kx - \omega t)} + D e^{-i(kx + \omega t)}$$

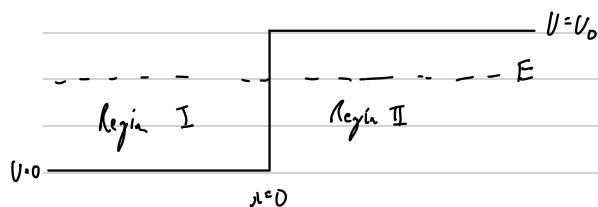
$C=0 \rightarrow$ left moving

$D=0 \rightarrow$ right moving

$C, D \neq 0 \rightarrow$ interference b/w opposite momenta

7.8 The Non-Rigid box

Potential Step



$x < 0$: Region I, $x > 0$: Region II! If $E > V_0$

$$\psi_1(x) = A e^{ik_1 x} + B e^{-ik_1 x} \quad \psi_2(x) = C e^{ik_2 x} \quad \text{since nothing is reflected from the right.}$$

$$k_2 = \sqrt{\frac{2m(E - V_0)}{\hbar^2}}$$

Boundary cond.

$$\psi_1(0) = \psi_2(0) \rightarrow A + B = C$$

$$\psi_1'(0) = \psi_2'(0) \rightarrow ik_1 A - ik_1 B = ik_2 C \rightarrow k_1 A - k_1 B = k_2 C$$

$$\rightarrow R = \left| \frac{B}{A} \right|^2 = \left(\frac{k_1 - k_2}{k_1 + k_2} \right)^2$$

$$T = \left| \frac{C}{A} \right|^2 = \left(\frac{2k_1}{k_1 + k_2} \right)^2$$

If $E < U_0$, Region II:

$$\psi_2(x) = C e^{-\alpha x} + D e^{+\alpha x}, \quad \alpha = \frac{\sqrt{2m(U_0 - E)}}{\hbar}$$

Boundary cond.

$$\psi_1(0) = \psi_2(0)$$

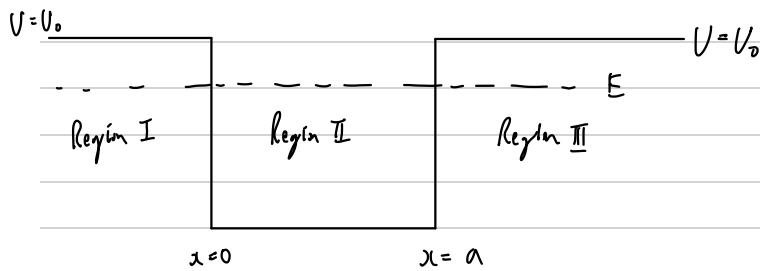
$$A + D = C$$

$$\psi_1'(0) = \psi_2'(0) \longrightarrow \frac{B}{A} = \frac{k_1 - i\alpha}{k_1 + i\alpha}$$

$$k_1(A - B) = i\alpha C$$

$$R = \left| \frac{B}{A} \right|^2 = 1$$

Finite potential well



Region II:

$$\psi_2(x) = C \sin(kx) + D \cos(kx), \quad k = \frac{\sqrt{2mE}}{\hbar}$$

Region I:

$$\psi_1(x) = A e^{-\alpha x} + B e^{+\alpha x}, \quad \alpha = \frac{\sqrt{2m(U_0 - E)}}{\hbar}$$

Region III:

$$\psi_3(x) = F e^{-\alpha x} + G e^{+\alpha x}, \quad \alpha = \frac{\sqrt{2m(U_0 - E)}}{\hbar}$$

Relativity (Classical)

Classically:

1. Newtonian mechanics holds in a family of reference frames where we move uniformly relative to any other
- (False) 2. There can be only one reference frame in which light travels at the same speed c in all directions

Postulates of Relativity

1. If S is an inertial frame and if a second frame S' moves with constant velocity relative to S , then S' is also an inertial frame
2. In all inertial frames, light travels through a vacuum with the same speed in any direction